

Use Case Description

Use Case name:	Get Directions to Parking Spot
Product name:	UrbanDrop
Team:	Fantastic4
Date:	2/24/26

1. Goal

Allow delivery personnel the ability to get to a communal parking spot, based on what parking spots are shown on the map.

2. Summary

The delivery personnel access the directions via the map interface. The system has already retrieved communal parking spots, and they are displayed on the map. The delivery personnel chooses a parking spot and receives directions from their location.

3. Actors

Actor 1: Delivery Personnel

4. Preconditions

- View Communal Parking Spots on map.
- View Parking Spot Details.
- The user is authenticated.
- The system is operational.
- The user's current location and parking spot location are available.

5. Trigger

The user selects "Get Directions" from the map interface via a button/icon.

6. Primary Sequence

Step	Action
1	The user sees available parking spots.

2	The user selects a specific parking spot.
3	The user selects the “get directions” icon.
4	The system visually shows directions from the user’s current location.

7. Primary Postconditions

- The user is able to view which specific parking spot they parked at.
- The user can save their spot by clicking a button so they don’t forget where they are parked.

8. Alternate Sequences

Alternate Trigger

Can’t navigate to the parking spot.

Step	Action
1	The system is not able to find directions to the specific parking spot.
2	The system says no path found.

Alternate Postconditions

- The map still shows the other parking spots.
 - The system suggests the next closest parking spot.
- etc.*

Alternate Trigger

Can’t receive user location.

Step	Action
1	The user presses the get directions button.
2	The system detects that the user location is unavailable.
3	The system sends an error message saying location not found.

Alternate Postconditions

- The system suggests restarting the app.
 - The system has a get help button to get help from a representative.
- etc.*

9. Nonfunctional Requirements

- The system shall continue to update parking spots every 2 seconds for 95% of the requests under normal conditions.
- If the directions button is unavailable the button should be greyed out once the user fails to get directions.

10. Glossary

- Communal Parking Spot: A designated legal short-term parking location for delivery personnel.
- Icon: A symbol or graphic representation on a screen of a program, option, or window, especially one of several for selection.

Noun–Class Table

Noun	Class Name	States	Member Variables
Delivery Personnel	User	authenticated / unauthenticated	userId, name, currentLocation, isAuthenticated
Parking Spot	ParkingSpot	available / unavailable / selected	spotId, coordinates, availabilityStatus, description
Map	MapView	loaded / not loaded	displayedSpots, userLocation
Direction	Route	generated / notFound	startLocation, endLocation, pathCoordinates, estimatedTime
Location	Location	valid / unavailable	latitude, longitude
System	UrbanDropSystem	operational / error	userList, parkingSpotList
Button/Icon	UIControl	enabled / disabled	controlId, label, state
Help Request	SupportRequest	open / resolved	requestId, userId, message

Verb–Behavior Table

Verb	Class Responsible	Member Function
View parking spots	MapView	displayParkingSpots()
Select parking spot	ParkingSpot	selectSpot()
Get directions	UrbanDropSystem	generateRoute(user, parkingSpot)
Display directions	MapView	displayRoute(route)
Save parking spot	User	saveParkingSpot(spot)
Update parking spots	UrbanDropSystem	refreshParkingSpots()
Authenticate user	User	authenticate(credentials)
Detect location	User	updateLocation()
Suggest parking spot	UrbanDropSystem	suggestNearestSpot(location)
Send error message	UrbanDropSystem	showError(message)